```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("/content/DataCo_Cleaned-2.csv", encoding='latin1')
```

```
df.head()
```

	Type	Days for shipping (real)	Days for shipment (scheduled)	Benefit per order	Sales per customer	Late_deliv
0	DEBIT	3	4	91.250000	314.640015	
1	TRANSFER	5	4	-249.089996	311.359985	
2	CASH	4	4	-247.779999	309.720001	
3	DEBIT	3	4	22.860001	304.809998	
4	PAYMENT	2	4	134.210007	298.250000	

5 rows × 35 columns

```
df.tail()
```

	Type	Days for shipping (real)	Days for shipment (scheduled)	Benefit per order	Sales per customer	Late_delive
59601	DEBIT	2	4	144.050003	293.980011	
59602	DEBIT	2	4	41.160000	293.980011	
59603	DEBIT	2	4	36.750000	293.980011	
59604	DEBIT	5	4	43.119999	391.980011	

```
print("Rows and Columns:", df.shape)
```

```
Rows and Columns: (59606, 35)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 59606 entries, 0 to 59605
Data columns (total 35 columns):
 #   Column                          Non-Null Count  Dtype  
---  ------                          --------------  -----  
 0   Type                            59606 non-null  object 
 1   Days for shipping (real)        59606 non-null  int64  
 2   Days for shipment (scheduled)   59606 non-null  int64  
 3   Benefit per order               59606 non-null  float64
 4   Sales per customer              59606 non-null  float64
 5   Late_delivery_risk              59605 non-null  float64
 6   Category Name                   59605 non-null  object 
 7   Customer City                   59605 non-null  object 
 8   Customer Country                59605 non-null  object 
 9   Customer Segment                59605 non-null  object 
 10  Customer State                  59605 non-null  object 
 11  Department Name                 59605 non-null  object 
 12  Latitude                        59605 non-null  float64
 13  Longitude                       59605 non-null  float64
 14  Market                          59605 non-null  object 
 15  Order City                      59605 non-null  object 
 16  Order Country                   59605 non-null  object 
 17  Order Item Discount             59605 non-null  float64
 18  Order Item Discount Rate        59605 non-null  float64
 19  Order Item Profit Ratio         59605 non-null  float64
 20  Order Item Quantity             59605 non-null  float64
 21  Sales                           59605 non-null  float64
 22  Order Item Total                59605 non-null  float64
 23  Order Profit Per Order          59605 non-null  float64
 24  Order Region                    59605 non-null  object 
 25  Order State                     59605 non-null  object 
 26  Order Status                    59605 non-null  object 
 27  Product Name                    59605 non-null  object 
 28  Product Price                   59605 non-null  float64
 29  Shipping Mode                   59605 non-null  object 
 30  order_month                     59605 non-null  float64
 31  order_day                       59605 non-null  float64
 32  order_weekday                   59605 non-null  float64
 33  order_hour                      59605 non-null  float64
 34  is_weekend                      59606 non-null  int64  
dtypes: float64(17), int64(3), object(15)
memory usage: 15.9+ MB
```

```
df.describe()
```

	Days for shipping (real)	Days for shipment (scheduled)	Benefit per order	Sales per customer	Late_delive
count	59606.000000	59606.000000	59606.000000	59606.000000	5960
mean	3.399423	2.800607	21.404607	175.577258	
std	1.634348	1.416627	102.681853	123.002609	
min	0.000000	0.000000	-3366.000000	7.490000	
25%	2.000000	2.000000	6.040000	99.989998	
50%	3.000000	4.000000	29.740000	151.160004	
75%	5.000000	4.000000	62.335000	239.979996	
max	6.000000	4.000000	864.000000	1799.989990	

```
df.isnull().sum()
```

	0
Type	0
Days for shipping (real)	0
Days for shipment (scheduled)	0
Benefit per order	0
Sales per customer	0
Late_delivery_risk	1
Category Name	1
Customer City	1
Customer Country	1
Customer Segment	1
Customer State	1
Department Name	1
Latitude	1
Longitude	1
Market	1
Order City	1

Order Country	1
Order Item Discount	1
Order Item Discount Rate	1
Order Item Profit Ratio	1
Order Item Quantity	1
Sales	1
Order Item Total	1
Order Profit Per Order	1
Order Region	1
Order State	1
Order Status	1
Product Name	1
Product Price	1
Shipping Mode	1
order_month	1
order_day	1
order_weekday	1
order_hour	1
is_weekend	0

dtype: int64

```
print("Duplicate Records:", df.duplicated().sum())
```

```
Duplicate Records: 0
```

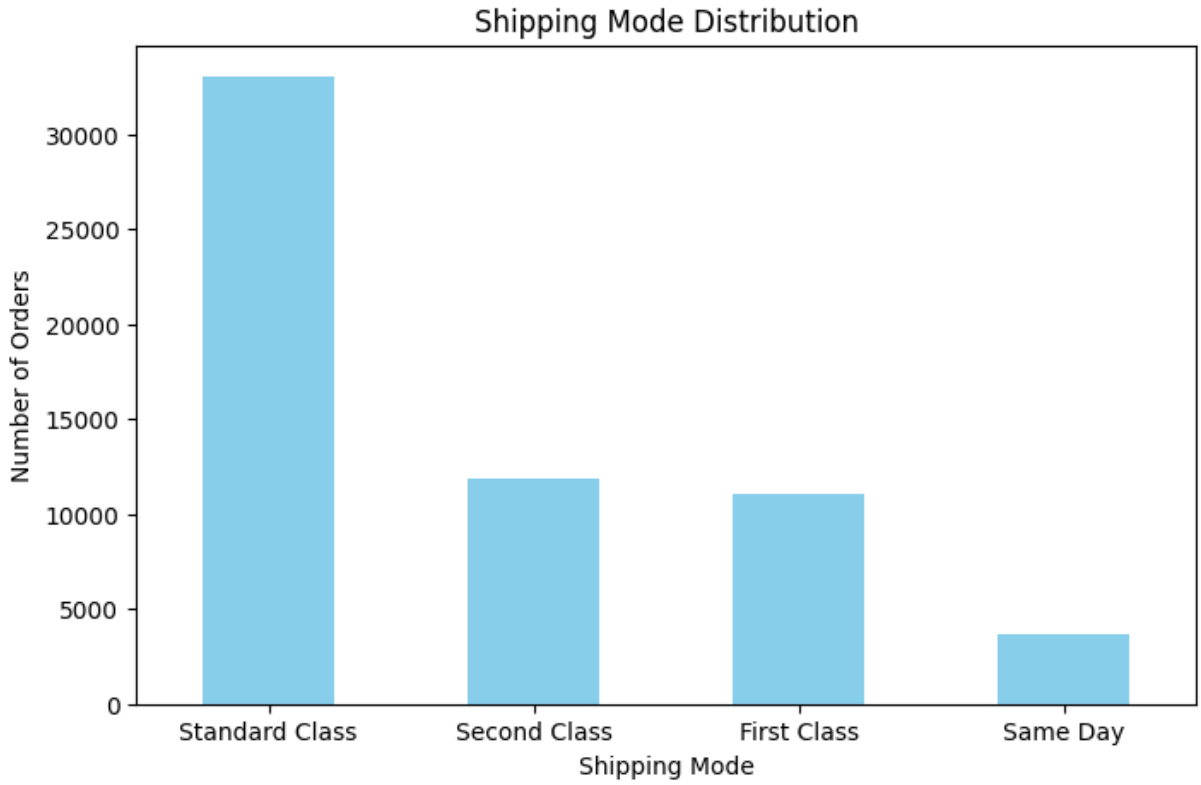
3.6 Pie Chart: Late Delivery Risk Distribution

```
shipping_mode = df['Shipping Mode'].value_counts()

plt.figure(figsize=(8,5))
shipping_mode.plot(kind='bar', color='skyblue')

plt.title("Shipping Mode Distribution")
plt.xlabel("Shipping Mode")
plt.ylabel("Number of Orders")
plt.xticks(rotation=0)

plt.show()
```



Purpose

This pie chart shows the percentage of orders having late delivery risk.

Observation

Standard Class has the highest number of orders.

Same Day has the lowest number of orders.

New Section

Customers mostly prefer economical shipping methods.

3.6 Pie Chart: Late Delivery Risk Distribution

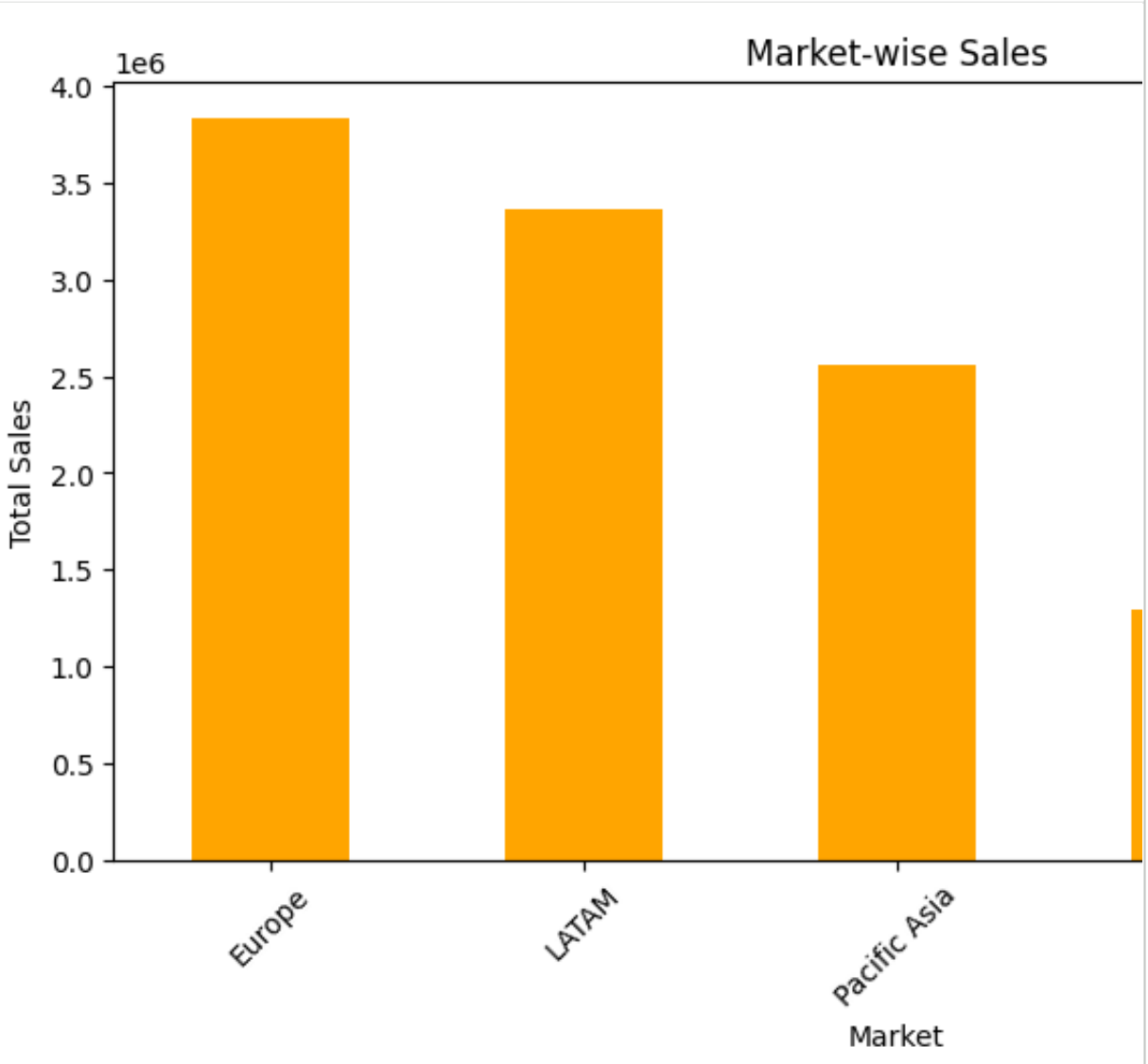
```
market_sales = df.groupby('Market')['Sales'].sum().sort_values(asce

plt.figure(figsize=(10,5))
market_sales.plot(kind='bar', color='orange')

plt.title("Market-wise Sales")
plt.xlabel("Market")
plt.ylabel("Total Sales")

plt.xticks(rotation=45)

plt.show()
```



Purpose

This visualization compares total sales across different markets. Observation

Displays markets generating the highest revenue.

Helps identify strong-performing markets.

3.8 Line Chart: Monthly Orders Trend

```
monthly_orders = df.groupby('order_month').size()

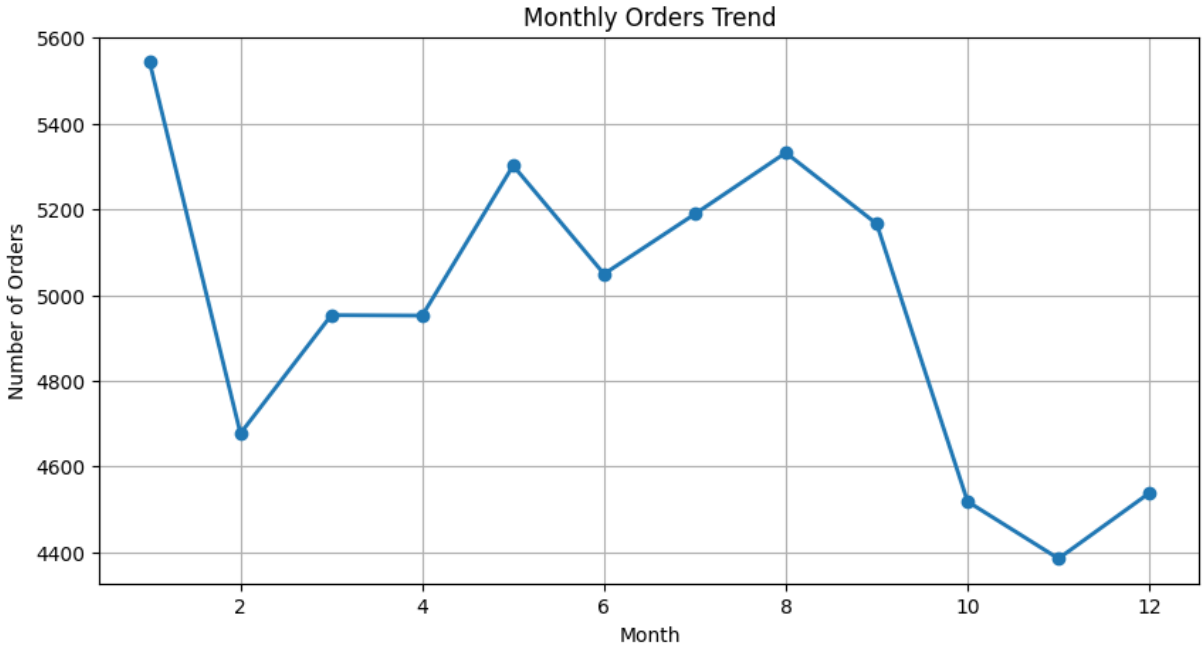
plt.figure(figsize=(10,5))

monthly_orders.plot(marker='o', linewidth=2)

plt.title("Monthly Orders Trend")
plt.xlabel("Month")
plt.ylabel("Number of Orders")

plt.grid(True)

plt.show()
```



Purpose

This line chart shows the monthly trend of orders.

Observation

Shows monthly increase or decrease in order volume.

Useful for identifying seasonal demand patterns.

3.9 Bar Chart: Product Category-wise Sales

```
# Product Category-wise Sales

category_sales = df.groupby('Category Name')['Sales'].sum().sort_va

plt.figure(figsize=(12,6))
category_sales.plot(kind='bar', color='green')

plt.title("Product Category-wise Sales")
plt.xlabel("Product Category")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)

plt.show()
```

Purpose

This bar chart compares the total sales generated by different product categories.

Observation

Displays sales generated by each product category.

Helps identify the best-selling product categories.

Higher bars indicate greater revenue contribution.

3.10 Grouped Bar Chart: Shipping Mode vs Late Delivery Risk

```
# Shipping Mode vs Late Delivery Risk

shipping_risk = pd.crosstab(df['Shipping Mode'],
                            df['Late_delivery_risk'])

shipping_risk.plot(kind='bar',
                   figsize=(8,5))

plt.title("Shipping Mode vs Late Delivery Risk")
plt.xlabel("Shipping Mode")
plt.ylabel("Number of Orders")
plt.legend(["No Risk","Risk"])

plt.show()
```



Purpose

This grouped bar chart compares late delivery risk across different shipping modes.

Observation

Compares delayed and on-time deliveries for each shipping mode.

Helps determine which shipping mode experiences more late deliveries.

Useful for logistics performance analysis.

3.11 Histogram: Distribution of Shipping Days

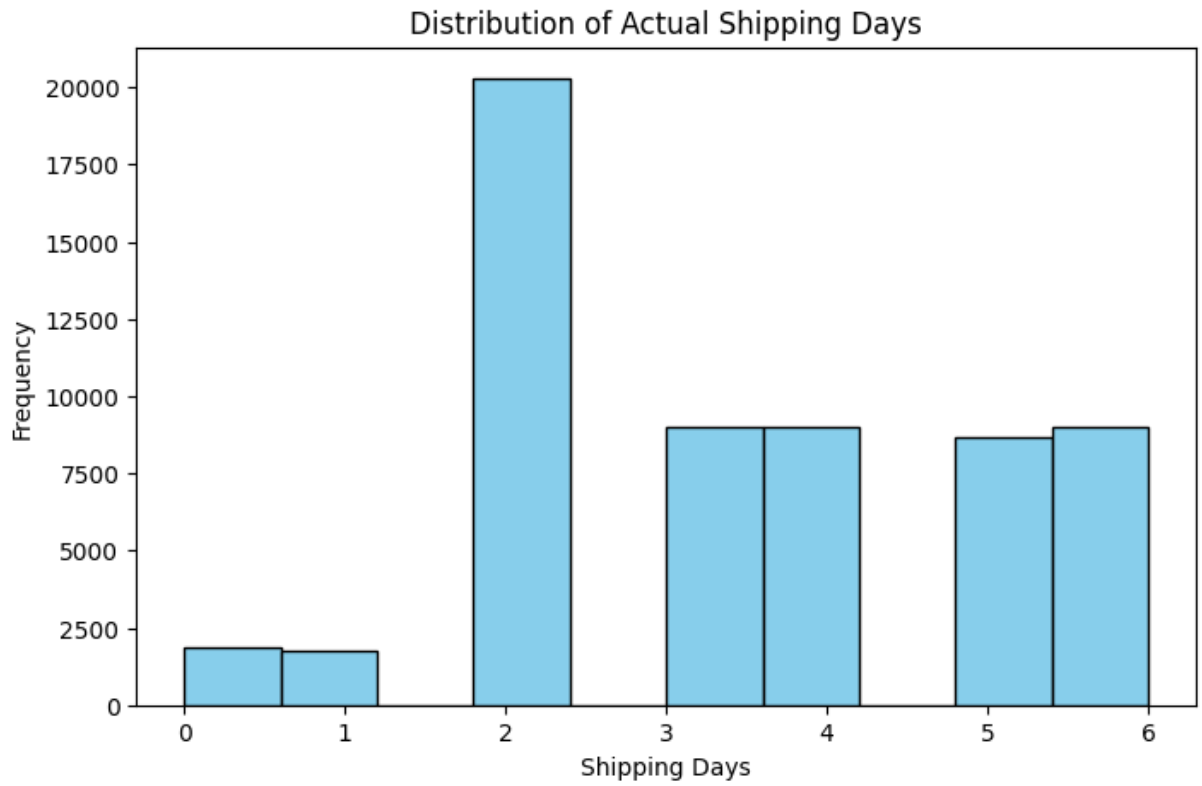
```
# Histogram of Actual Shipping Days

plt.figure(figsize=(8,5))

plt.hist(df['Days for shipping (real)'],
         bins=10,
         color='skyblue',
         edgecolor='black')

plt.title("Distribution of Actual Shipping Days")
plt.xlabel("Shipping Days")
plt.ylabel("Frequency")

plt.show()
```



3.12 Box Plot: Shipping Days

```
# Box Plot of Shipping Days

plt.figure(figsize=(6,5))

plt.boxplot(df['Days for shipping (real)'])

plt.title("Box Plot of Actual Shipping Days")
plt.ylabel("Shipping Days")

plt.show()
```



Observation Displays minimum, maximum, median, and quartiles.

Detects unusually long or short shipping durations (outliers).

Helps evaluate the consistency of delivery times.

3.13 Scatter Plot: Sales vs Actual Shipping Days

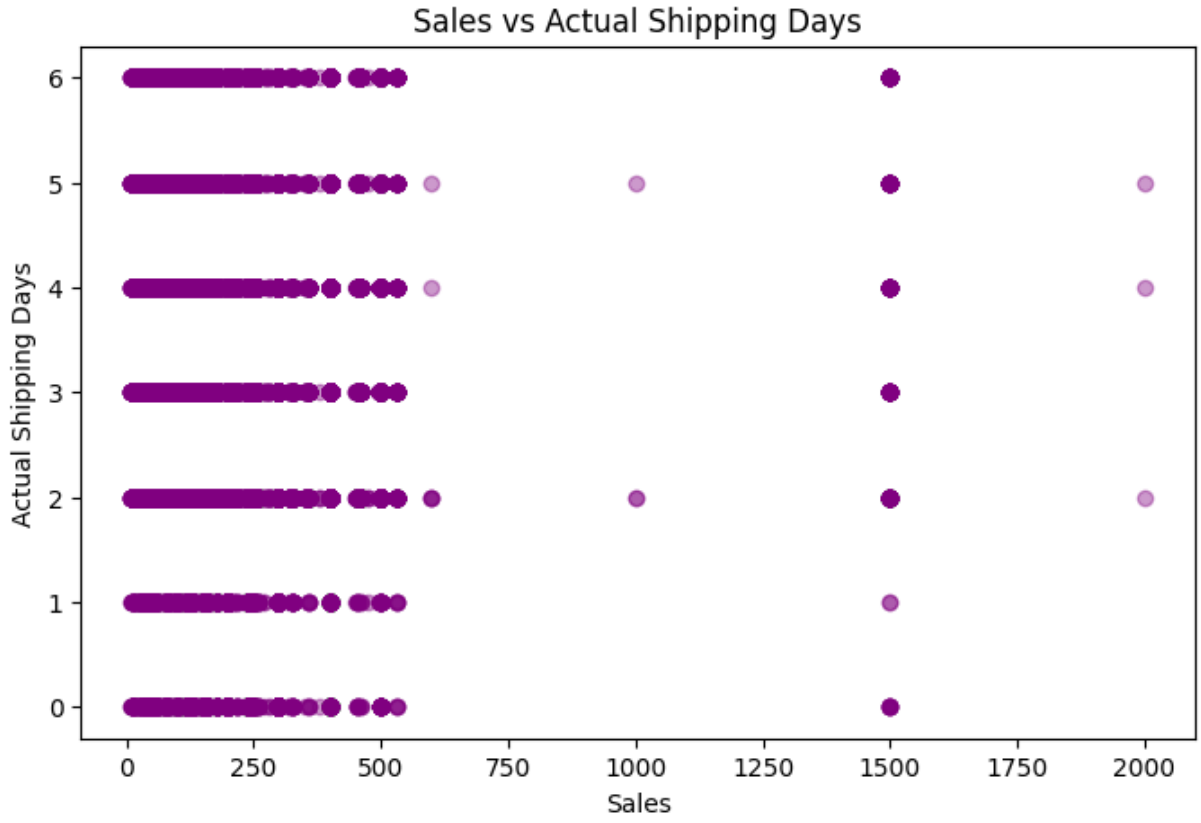
```
# Scatter Plot: Sales vs Actual Shipping Days

plt.figure(figsize=(8,5))

plt.scatter(df['Sales'],
            df['Days for shipping (real)'],
            alpha=0.4,
            color='purple')

plt.title("Sales vs Actual Shipping Days")
plt.xlabel("Sales")
plt.ylabel("Actual Shipping Days")

plt.show()
```



Observation Shows the relationship between sales amount and shipping days.

Helps identify whether higher sales orders require longer delivery times.

Useful for detecting patterns or outliers.

3.14 Customer-Wise Purchase Analysis

```
# Customer-Wise Purchase Analysis

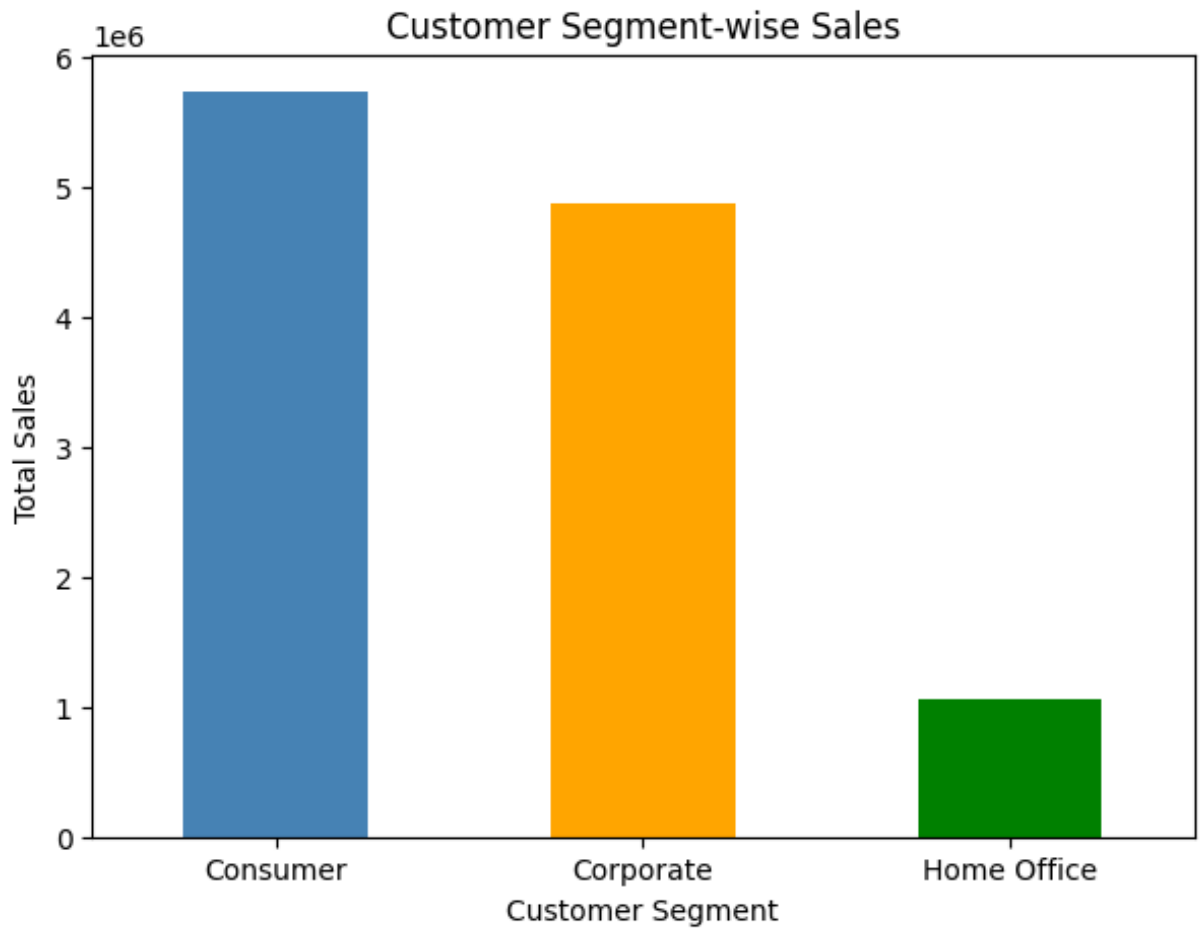
customer_sales = df.groupby('Customer Segment')['Sales'].sum()

plt.figure(figsize=(7,5))

customer_sales.plot(kind='bar',
                    color=['steelblue','orange','green'])

plt.title("Customer Segment-wise Sales")
plt.xlabel("Customer Segment")
plt.ylabel("Total Sales")
plt.xticks(rotation=0)

plt.show()
```



3.15 Heatmap: Correlation Between Numerical Attributes

```
# Heatmap: Correlation Between Numerical Attributes

import matplotlib.pyplot as plt

corr = df[['Sales',
```

```
            'Benefit per order',
            'Order Item Quantity',
            'Days for shipping (real)',
            'Days for shipment (scheduled)',
            'Product Price']].corr()

plt.figure(figsize=(8,6))

plt.imshow(corr,
           cmap='coolwarm',
           interpolation='nearest')

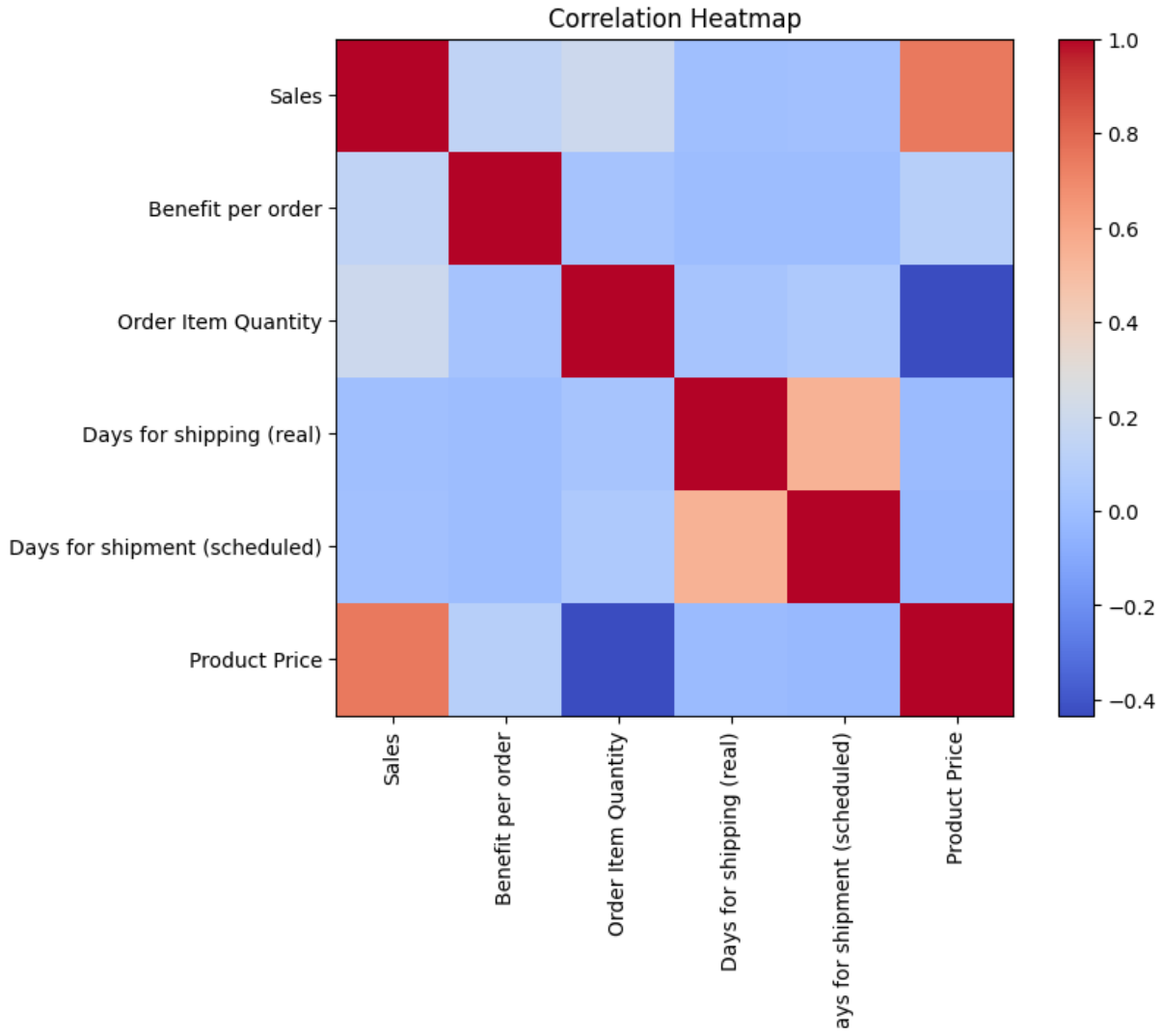
plt.colorbar()

plt.xticks(range(len(corr.columns)),
           corr.columns,
           rotation=90)

plt.yticks(range(len(corr.columns)),
           corr.columns)

plt.title("Correlation Heatmap")

plt.show()
```



Observation

Shows positive and negative correlations between numerical attributes.

Stronger colours indicate stronger relationships.

Helps identify variables useful for shipping days prediction.

3.16 Monthly Sales Visualization

Purpose

This visualization shows total sales across different months.

```
# Monthly Sales Visualization

monthly_sales = df.groupby('order_month')['Sales'].sum()

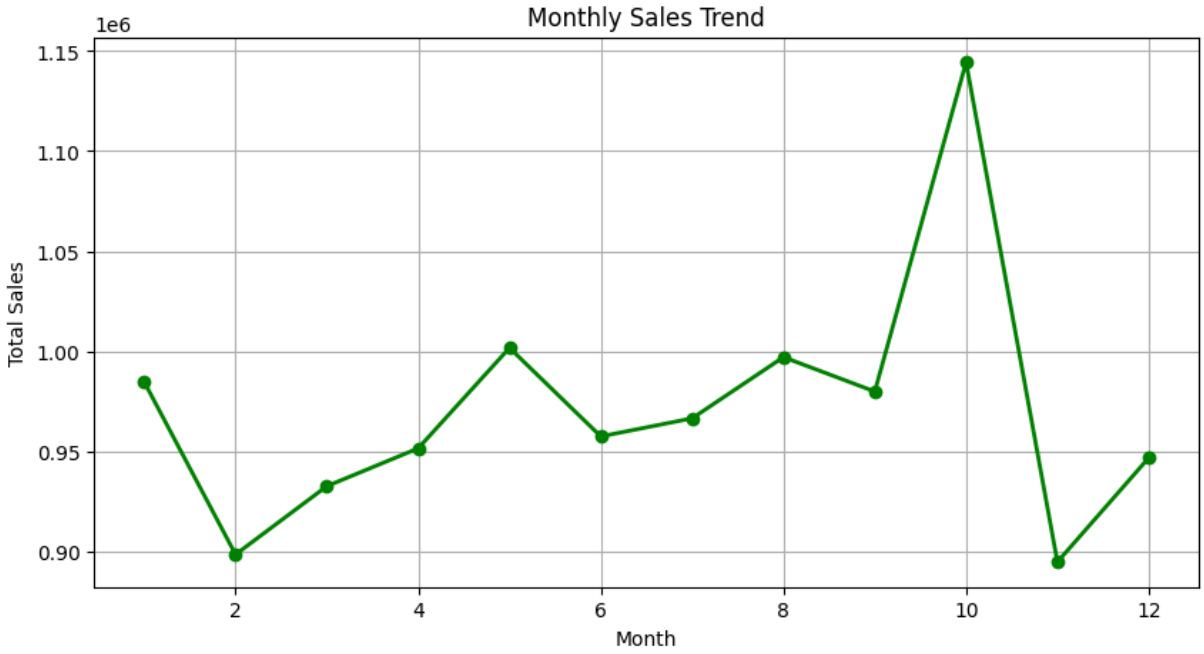
plt.figure(figsize=(10,5))

monthly_sales.plot(marker='o',
                   linewidth=2,
                   color='green')

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")

plt.grid(True)

plt.show()
```



3.17 Dashboard-Style Combined Analysis

Purpose

This dashboard combines four important visualizations in one figure to provide an overall view of shipping and sales performance.

```
# Dashboard-Style Combined Analysis

fig, axes = plt.subplots(2, 2, figsize=(14,10))

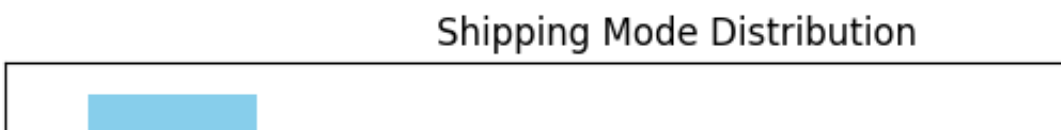
# Shipping Mode Distribution
df['Shipping Mode'].value_counts().plot(
    kind='bar',
    ax=axes[0,0],
    color='skyblue'
)
axes[0,0].set_title("Shipping Mode Distribution")

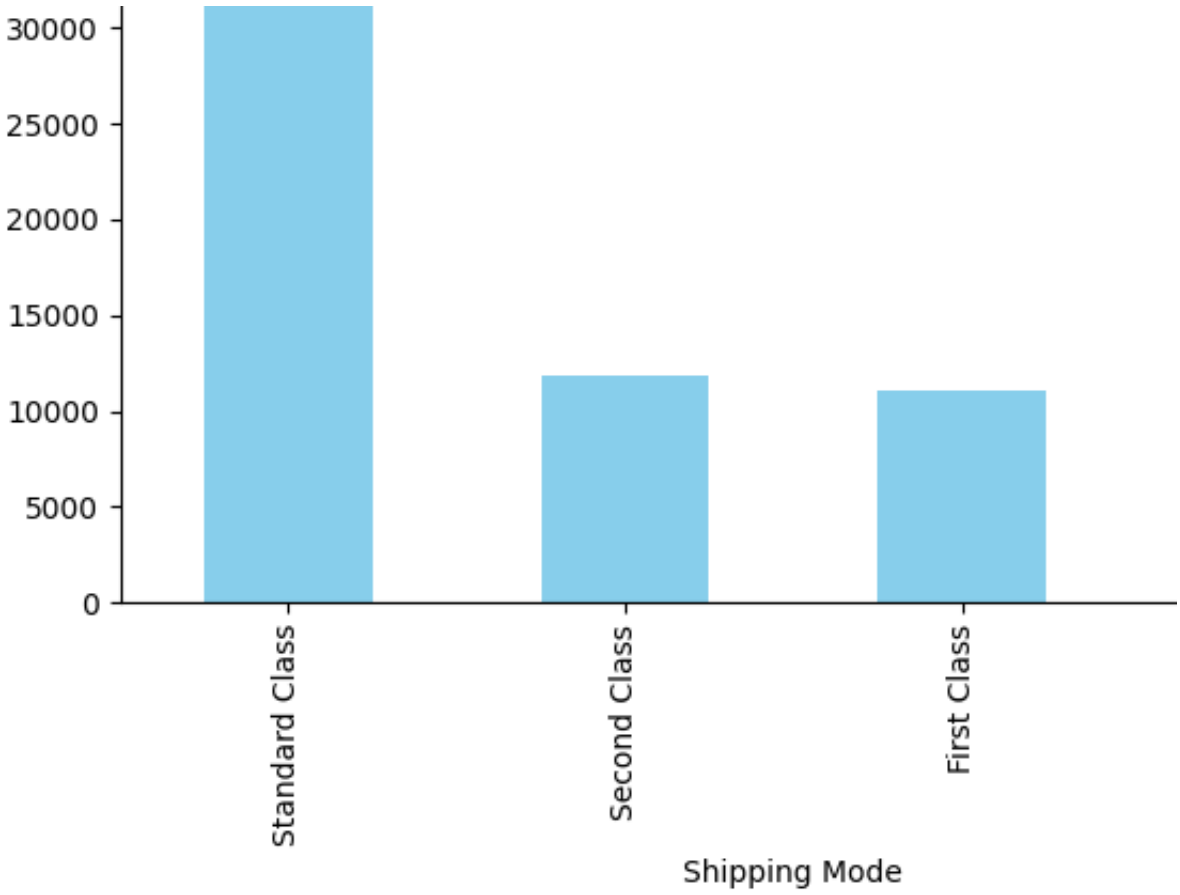
# Monthly Sales Trend
df.groupby('order_month')['Sales'].sum().plot(
    ax=axes[0,1],
    marker='o',
    color='green'
)
axes[0,1].set_title("Monthly Sales Trend")

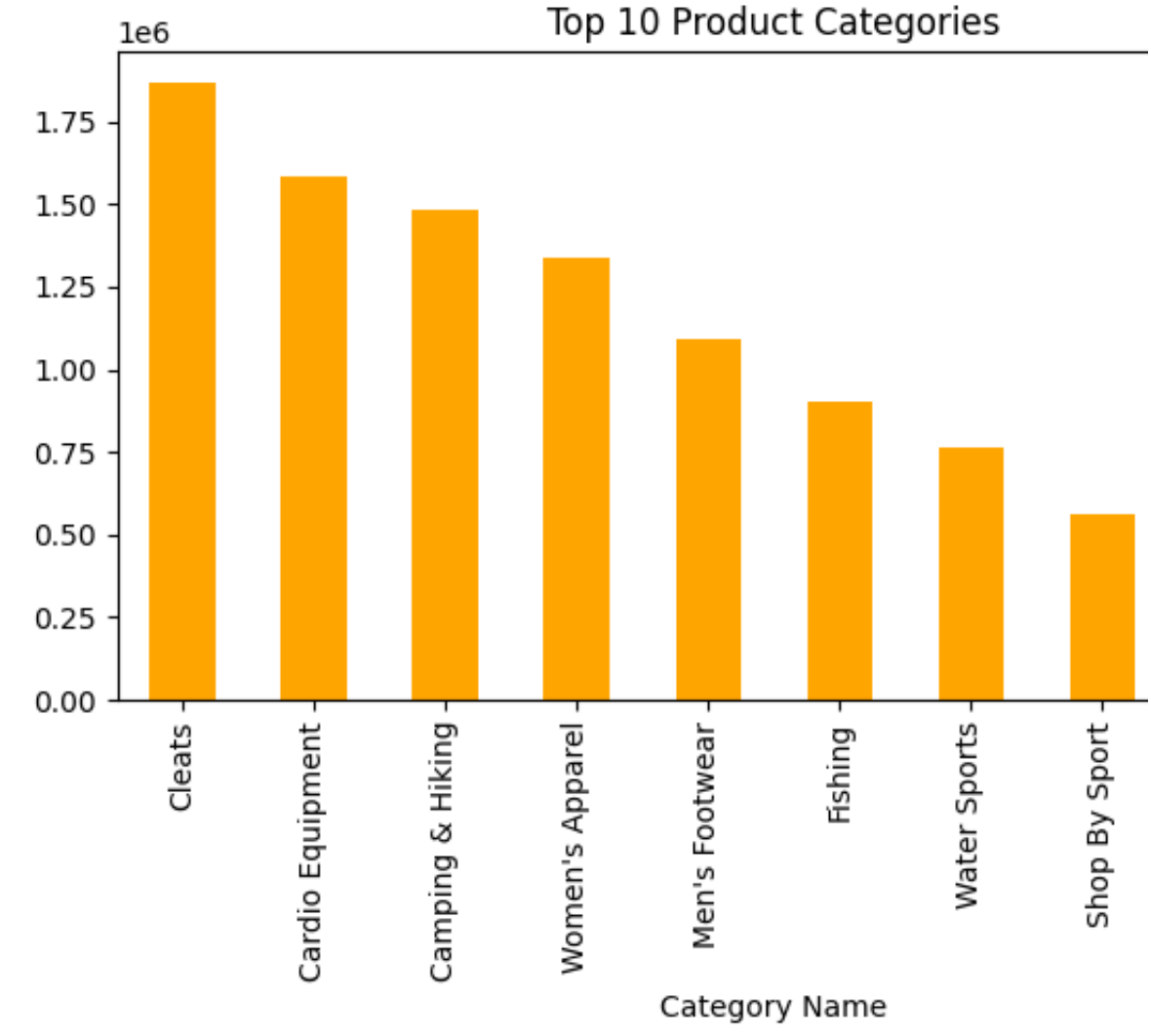
# Category-wise Sales
df.groupby('Category Name')['Sales'].sum().sort_values(ascending=Fal
    kind='bar',
    ax=axes[1,0],
    color='orange'
)
axes[1,0].set_title("Top 10 Product Categories")

# Late Delivery Risk
df['Late_delivery_risk'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%',
    ax=axes[1,1]
)
axes[1,1].set_title("Late Delivery Risk")
axes[1,1].set_ylabel("")

plt.tight_layout()
plt.show()
```




30000
25000
20000
15000
10000
5000
0
Standard Class
Second Class
First Class
Shipping Mode


Top 10 Product Categories
1e6
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Cleats
Cardio Equipment
Camping & Hiking
Women's Apparel
Men's Footwear
Fishing
Water Sports
Shop By Sport
Category Name

Observation

The dashboard helps management identify:

Most preferred shipping mode.

Monthly sales performance.

Best-selling product categories.

Percentage of late deliveries.

Overall supply chain performance in a single view.